

Trace extensions and faithful representations on a separable space

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Scope

The CAR trace extends uniquely to a state on the same shell-generated target, and that extension is tracial. Pointed transport supplies faithful representations on separable trace spaces; representation faithfulness is distinct from faithfulness of a vector state.

9 direct Cards + 30 reused prerequisites = 39 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Uniqueness among all state extensions of the CAR trace
- Key result: Faithfulness of the representation on the CAR trace Hilbert space

[Read this route with prerequisites](#)

Detailed route conditions

Fix the same T and ι , and first extend the CAR trace τ to a state ϕ on T . Existence precedes uniqueness and traciality. In each trace-realizing cyclic representation, the estimates for f_n and $\alpha_j(f_n)$ kill the residual projections on the source-cyclic subspace. Target cyclicity then proves source cyclicity. The fixed-left-hand-side limit calculation is retained with its full n dependence in the pointed-transport Card.

Pointed transport proves uniqueness among all state extensions. Uniqueness gives invariance under source unitaries; their span and the actual represented shell limits put all generators in the closed centralizer. This proves traciality. A different restriction argument then proves uniqueness among all tracial states.

The CAR trace space and the extension-state space are different: the selected unitary points as $E : H_\tau \rightarrow H_\phi$, and $\tilde{\rho}(t) = E^{-1}\rho_\phi(t)E$. This E is not the capture map W . Faithfulness of a representation is distinct from faithfulness of its vector state. The Hilbert spaces are separable here; norm separability of T is not assumed.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (5 Cards)

Level 0

The selected GNS representation of a pure-state class

A choice of representative pure states, fixed literally at a root, gives one concrete GNS representation per equivalence class.

`MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations

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Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra

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Level 0

A fixed representative family of CAR shell data

One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component.

`MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · Shell matching fixes the trace of transported flags · An inequivalent GNS fiber has no residual common range · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation

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Level 0 The closed operator algebra generated by a representation and extra operators Places a represented algebra and a chosen family of bounded operators in one concrete closed algebra. MathlibAnnex.CStarAlgebra.AtomicConstruction.concreteTarget Immediate Card prerequisites: None in this selected scope Used by in this scope: An irreducible operator algebra constructed from projection shells · Reconstructing a represented unitary from its shells and residual corner Open Declaration Card Read exact source
Level 0 The normalized trace on a finite CAR stage The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions. MathlibAnnex.CStarAlgebra.CAR.stageTrace Immediate Card prerequisites: None in this selected scope Used by in this scope: Constructing the normalized trace on the completed CAR algebra Open Declaration Card Read exact source
Level 1 (5 Cards)
Level 1 Constructing the normalized trace on the completed CAR algebra Carries compatible bounded matrix traces through the algebraic limit, normed union and completion. MathlibAnnex.CStarAlgebra.CAR.trace Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages Used by in this scope: Shell matching fixes the trace of transported flags · Extending the CAR trace to the fixed shell target · Normalization and the trace identity determine the CAR trace Open Declaration Card Read exact source
Level 1 The product-vector state on the completed CAR algebra Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion. MathlibAnnex.CStarAlgebra.CAR.rootState Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state Open Declaration Card Read exact source

Level 1 An irreducible operator algebra constructed from projection shells Shell partial isometries with rank-one limiting defects can be completed to unitary links that join inequivalent irreducible fibers. MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_atomicShellModel
Immediate Card prerequisites: The closed operator algebra generated by a representation and extra operators Used by in this scope: The atomic-shell construction on selected pure-GNS fibers
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Level 1 Distinct GNS classes have no unitary intertwiner The class index makes the selected pure-GNS family pairwise unitarily inequivalent. MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation
Immediate Card prerequisites: The selected GNS representation of a pure-state class Used by in this scope: An inequivalent GNS fiber has no residual common range · The atomic-shell construction on selected pure-GNS fibers · Assembling selected GNS cyclic subspaces into an isometric source representation
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Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm · Shell matching fixes the trace of transported flags · Reconstructing a represented unitary from its shells and residual corner
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Level 2 (5 Cards)

Level 2

Root compression becomes scalar in norm

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range · A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

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Level 2

Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: The matching GNS fiber retains exactly its cyclic line · The atomic direct sum of the selected CAR representations · An inequivalent GNS fiber has no residual common range

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Level 2

Normalization and the trace identity determine the CAR trace

Proves uniqueness without assuming positivity of the competing continuous functional.

`MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm`

Immediate Card prerequisites: Constructing the normalized trace on the completed CAR algebra

Used by in this scope: The fixed shell target has a unique tracial state

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Level 2 The atomic-shell construction on selected pure-GNS fibers Pure-state GNS data supplies the irreducible and inequivalent fibers required by the generic shell construction. MathlibAnnex.CStarAlgebra.AtomicConstruction.exists_irreducible_pureAtomicShellModel Immediate Card prerequisites: Distinct GNS classes have no unitary intertwiner · An irreducible operator algebra constructed from projection shells Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra Open Declaration Card Read exact source
Level 2 Shell matching fixes the trace of transported flags Recovers exact trace values from the initial and final supports of shell links. MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag Immediate Card prerequisites: The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · A fixed representative family of CAR shell data Used by in this scope: Transported flags vanish on vectors generated by a trace vector Open Declaration Card Read exact source
Level 3 (4 Cards) Level 3 The atomic direct sum of the selected CAR representations All selected pure-GNS fibers of the completed CAR algebra act together on one Hilbert direct sum. MathlibAnnex.CStarAlgebra.CAR.selectedAtomicRepresentation Immediate Card prerequisites: The selected GNS representation of a pure-state class · Purity of the completed root state Used by in this scope: Realizing a shell family by a faithful irreducible operator algebra · Reconstructing a represented unitary from its shells and residual corner · Assembling selected GNS cyclic subspaces into an isometric source representation Open Declaration Card Read exact source
Level 3 The matching GNS fiber retains exactly its cyclic line Identifies the residual projection of a transported CAR flag in its matching pure-state representation. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_rankOne Immediate Card prerequisites: Root compression becomes scalar in norm · The selected GNS representation of a pure-state class · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source

Level 3 Transported flags vanish on vectors generated by a trace vector Turns decay of projection traces into norm convergence on each source-orbit vector. MathlibAnnex.CStarAlgebra.CAR.tendsto_transportedFlag_orbit_zero Immediate Card prerequisites: Shell matching fixes the trace of transported flags Used by in this scope: The target represented on the CAR trace GNS space · Pointed unitary transport between trace-cyclic target representations Open Declaration Card Read exact source
Level 3 An inequivalent GNS fiber has no residual common range Uses compression and cyclic transport to exclude fixed vectors in every other chosen pure-state class. MathlibAnnex.CStarAlgebra.CAR.selected_commonFixedProjection_eq_zero Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · Purity of the completed root state · A fixed representative family of CAR shell data Used by in this scope: The atomic common range is one embedded GNS line Open Declaration Card Read exact source
Level 4 (3 Cards)
Level 4 The atomic common range is one embedded GNS line Assembles the matching and inequivalent fiber calculations in an arbitrary Hilbert direct sum. MathlibAnnex.CStarAlgebra.CAR.iInf_range_atomic_transportedFlag_eq_span Immediate Card prerequisites: The matching GNS fiber retains exactly its cyclic line · An inequivalent GNS fiber has no residual common range Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion · Realizing a shell family by a faithful irreducible operator algebra Open Declaration Card Read exact source
Level 4 Assembling selected GNS cyclic subspaces into an isometric source representation Joins mutually orthogonal pure-state cyclic copies and controls the limiting flag projections on their sum. MathlibAnnex.CStarAlgebra.CAR.exists_selectedAtomicCyclicIsometry Immediate Card prerequisites: Root compression becomes scalar in norm · Distinct GNS classes have no unitary intertwiner · The atomic direct sum of the selected CAR representations · A fixed representative family of CAR shell data Used by in this scope: The cyclic sum fills every irreducible target representation Open Declaration Card Read exact source

Level 4 Reconstructing a represented unitary from its shells and residual corner Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces. <code>MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction</code> Immediate Card prerequisites: The atomic direct sum of the selected CAR representations · The decreasing root projections in the CAR completion · A fixed representative family of CAR shell data · The closed operator algebra generated by a representation and extra operators Used by in this scope: The target represented on the CAR trace GNS space · Pointed unitary transport between trace-cyclic target representations · An irreducible target representation has a surviving fixed space Open Declaration Card Read exact source
Level 5 (2 Cards)
Level 5 Realizing a shell family by a faithful irreducible operator algebra Constructs unitary links between pure-state summands and an irreducible algebra containing a faithful copy of CAR. <code>MathlibAnnex.CStarAlgebra.CAR.exists_completedAtomicShellModel</code> Immediate Card prerequisites: The atomic common range is one embedded GNS line · The atomic direct sum of the selected CAR representations · The atomic-shell construction on selected pure-GNS fibers Used by in this scope: The ambient inclusion of the fixed shell-family target · The CAR source map into the fixed shell-family target Open Declaration Card Read exact source
Level 5 An irreducible target representation has a surviving fixed space Rules out simultaneous disappearance of all limiting CAR flags by passing reduction through the reconstructed generators. <code>MathlibAnnex.CStarAlgebra.CAR.exists_nonzero_targetFixedSpace</code> Immediate Card prerequisites: Reconstructing a represented unitary from its shells and residual corner Used by in this scope: A common root vector realizes every selected state through the generators Open Declaration Card Read exact source

Level 6 (3 Cards)

Level 6 A common root vector realizes every selected state through the generators Builds a compatible family of unit vectors from a surviving residual space and identifies their CAR vector states. MathlibAnnex.CStarAlgebra.CAR.exists_targetDefectVectorFamily
Immediate Card prerequisites: Root compression becomes scalar in norm · An irreducible target representation has a surviving fixed space Used by in this scope: The cyclic sum fills every irreducible target representation
Open Declaration Card Read exact source
Level 6 The CAR source map into the fixed shell-family target Restricting the codomain of the selected atomic representation gives the source map into one fixed generated target. MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom
Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: Closed-ideal simplicity of the shell-generated algebra · Pointed unitary transport between trace-cyclic target representations · Extending the CAR trace to the fixed shell target
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Level 6 The ambient inclusion of the fixed shell-family target The same concrete target has a literal representation on the selected atomic Hilbert space. MathlibAnnex.CStarAlgebra.CAR.shellFamilyInclusion
Immediate Card prerequisites: Realizing a shell family by a faithful irreducible operator algebra Used by in this scope: Closed-ideal simplicity of the shell-generated algebra
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Level 7 (3 Cards)

Level 7

Pointed unitary transport between trace-cyclic target representations

Source cyclicity and strong shell sums extend the canonical CAR orbit isometry to the whole target.

`MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic`

Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: The unique trace extension is tracial on the whole target · Uniqueness among all state extensions of the CAR trace

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Level 7

Extending the CAR trace to the fixed shell target

The source trace first becomes a state on the existing target algebra.

`MathlibAnnex.CStarAlgebra.CAR.exists_state_extension_trace`

Immediate Card prerequisites: The CAR source map into the fixed shell-family target · Constructing the normalized trace on the completed CAR algebra

Used by in this scope: The GNS representation of the target trace extension

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Level 7

The cyclic sum fills every irreducible target representation

Uses shell reconstruction and residual lines to turn an isometric CAR intertwiner into a unitary equivalence on the source.

`MathlibAnnex.CStarAlgebra.CAR.exists_surjective_selectedAtomicCyclicIsometry`

Immediate Card prerequisites: A common root vector realizes every selected state through the generators · Assembling selected GNS cyclic subspaces into an isometric source representation

Used by in this scope: Every irreducible representation is unitarily equivalent to the inclusion

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Level 8 (2 Cards)

Level 8

The GNS representation of the target trace extension

The chosen state on the existing target supplies its cyclic GNS representation.

`MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation`

Immediate Card prerequisites: Extending the CAR trace to the fixed shell target

Used by in this scope: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation · The unique trace extension is tracial on the whole target · Uniqueness among all state extensions of the CAR trace

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Level 8

Every irreducible representation is unitarily equivalent to the inclusion

Extends the unitary equivalence from the CAR source to all additional generators and the norm-closed target.

`MathlibAnnex.CStarAlgebra.CAR.ambientInclusion_unitaryEquivalent`

Immediate Card prerequisites: The atomic common range is one embedded GNS line · The cyclic sum fills every irreducible target representation

Used by in this scope: Closed-ideal simplicity of the shell-generated algebra

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Level 9 (4 Cards)

Level 9

The target represented on the CAR trace GNS space

A fixed pointed unitary transports the target action onto the Hilbert space constructed from CAR alone.

`MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation`

Immediate Card prerequisites: The GNS representation of the target trace extension · Transported flags vanish on vectors generated by a trace vector · Reconstructing a represented unitary from its shells and residual corner

Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space

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Level 9

Uniqueness among all state extensions of the CAR trace

Pointed transport identifies arbitrary state extensions before traciality is proved.

`MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace`

Immediate Card prerequisites: The GNS representation of the target trace extension · Pointed unitary transport between trace-cyclic target representations

Used by in this scope: None in this selected scope

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Level 9 The unique trace extension is tracial on the whole target Unitary invariance, strong shell sums, and a closed centralizer extend the trace identity beyond the source. MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm Immediate Card prerequisites: The GNS representation of the target trace extension · Pointed unitary transport between trace-cyclic target representations Used by in this scope: The fixed shell target has a unique tracial state Open Declaration Card Read exact source
Level 9 Closed-ideal simplicity of the shell-generated algebra Uses a faithful irreducible representation in the unique unitary-equivalence class to rule out proper nonzero closed two-sided ideals. MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_shellFamilyTarget Immediate Card prerequisites: The ambient inclusion of the fixed shell-family target · Every irreducible representation is unitarily equivalent to the inclusion · The CAR source map into the fixed shell-family target Used by in this scope: Faithfulness of the target-state GNS representation Open Declaration Card Read exact source
Level 10 (2 Cards) Level 10 The fixed shell target has a unique tracial state Every tracial state is forced back to the CAR trace and hence to its unique extension. MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state Immediate Card prerequisites: Normalization and the trace identity determine the CAR trace · The unique trace extension is tracial on the whole target Used by in this scope: None in this selected scope Open Declaration Card Read exact source
Level 10 Faithfulness of the target-state GNS representation The ideal dichotomy of the same target forces the constructed nonzero representation to be injective. MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective Immediate Card prerequisites: The GNS representation of the target trace extension · Closed-ideal simplicity of the shell-generated algebra Used by in this scope: Faithfulness of the representation on the CAR trace Hilbert space Open Declaration Card Read exact source

Level 11 (1 Card)

Level 11 Faithfulness of the representation on the CAR trace Hilbert space Unitary conjugation preserves injectivity of the target-state representation. MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective
Immediate Card prerequisites: The target represented on the CAR trace GNS space · Faithfulness of the target-state GNS representation Used by in this scope: None in this selected scope
Open Declaration Card Read exact source